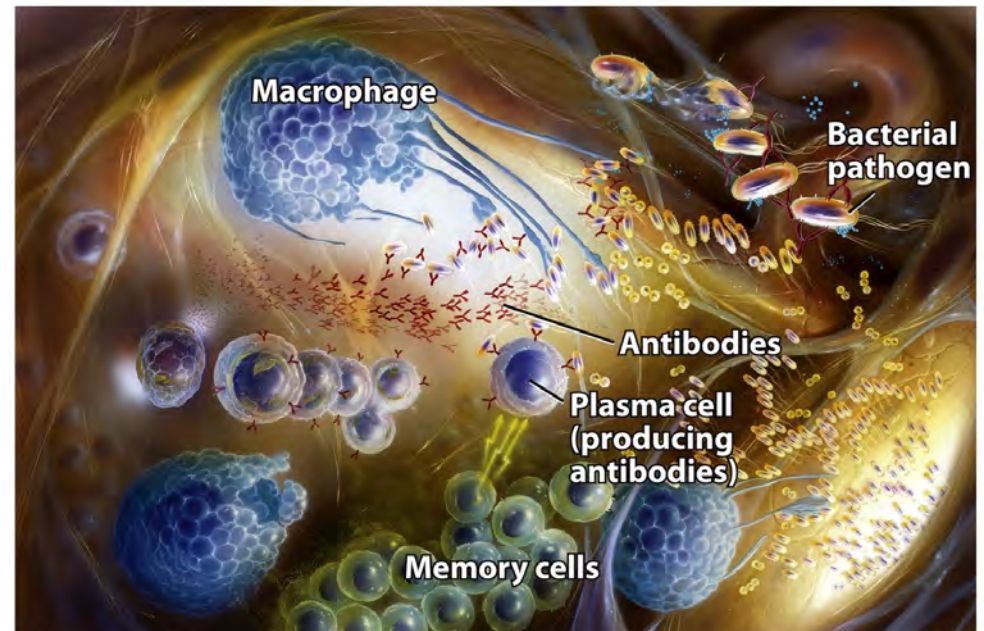
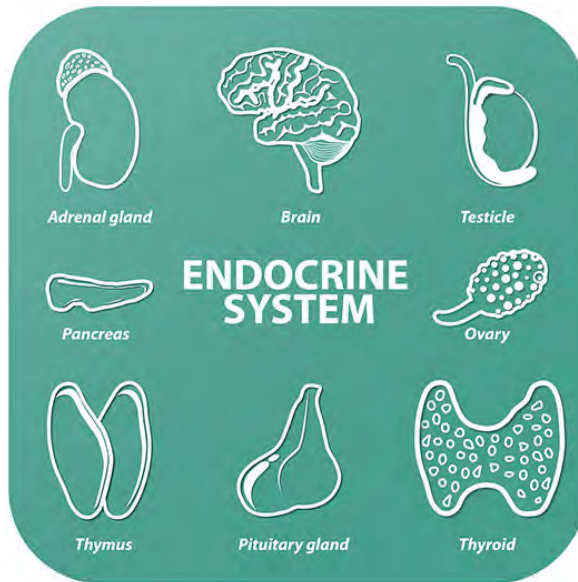


Endocrine System & Bodily Defenses

Instructor – Jeff Schinske



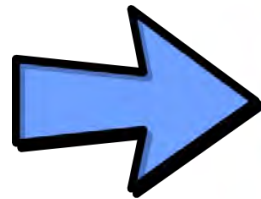


Bio 40C Proposed Norms

- Seek to enjoy your colleagues and your experiences in class!
- Actively select language that is inclusive, respectful, and welcoming
- Assume good intentions...no heroes and no villains (in this class)
- Recognize that we all have strengths as well as areas for growth
- Be curious, be skeptical
- Ask for help (and expect to receive it!)

Share with a partner...

1. Name
2. Something important to know about you as a person
3. What most stood out to you from the last week of Bio 40C.



Person with 1st name closest to start of alphabet goes first



Debriefing Last Week

○ Anonymous End of Class Index Cards

- "I didn't know natural killer cells existed so it's pretty cool to learn about them! I think I would also want to be an NK cell if I had to choose a cell."
- "Today I learned that much of what I thought was negative about getting sick like fevers was actually my body trying to defend me."
- "I enjoy these labs where we learn by looking at changes in our own bodies."

○ Reflecting On Canvas

- "I really enjoyed the hormone cases. It was fun trying to figure out what the patients condition was based off of their symptoms and test results, I felt like I was in an episode of Grey's Anatomy."
- "I enjoy seeing what my classmate thought looked wrong and compared to what I found. I also enjoy doing the patient introduction, Interpreting Test Result with classmates and lastly Checking Understanding quiz."



A Brief Preview of the Week

- Today

- Extending our work on the hormone case studies.
- Considering the adaptive immune responses using an immune system case study.

- Lab This Week

- Immune System

- Online Lecture Work This Week

- Introducing our last organ system!

A reminder of our first hormone case study...

Daisee is a 67-year old female who comes into your clinic as a new patient. You notice from the weight and height information on the chart that she is overweight, and you make a note to ask her about her level of physical activity and diet. Daisee says she's mostly gotten through life pretty healthy, but she's been annoyed recently by some minor symptoms. She feels thirsty all the time, and also uses the bathroom a lot. She says she finds this confusing. "Why do I feel so thirsty if I have to use the restroom all the time?" she asks. "My body tells me it wants the water, but then it gets rid of it!"



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In the space provided, create a couple rough diagrams...

- 1) A diagram of how insulin normally acts on a target cell to allow sugars to enter
- 2) A diagram of what you think is happening instead in Daisee's cells



A reminder about our second hormone case study...

Jordan is a 33-year old male who visits the doctor after feeling very tired and weak for an extended period of time. He says he often feels dizzy when he stands up, and sometimes feels nauseous. These symptoms have caused him quite a bit of anxiety, as he's worried that something serious is wrong.





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In the space provided, create a couple rough diagrams...

- 1) A diagram of how cortisol normally acts on a target cell to reduce inflammation
- 2) A diagram of what you think is happening instead in Jordan's body that might cause increased inflammatory responses in his body



An A&P Perspective On Not Touching Our Faces



We have all likely heard that we should wash our hands and avoid touching our faces with dirty hands. But digging into the A&P (cell/chemical) perspective of this, why is that? And what would proceed to happen if you do touch your face with virus particles on your hands?

Discuss at least four nonspecific/innate defenses with at least one of those four defenses involving a white blood cell of innate immunity.

Case 1: Xirui



Xirui, a 34 year old female, is at the doctor primarily due to joint pain. As you chat with Xirui, she explains, “For a couple months now I’ve been getting these occasional ‘attacks’ of joint pain. I remember one time I drove home from school and then the joint pain came on. I could barely move my wrists and shoulders, and had to sit in my car waiting for the pain to go away. It kept on, so I had to very slowly make my way from the garage to my apartment, trying not to extend my arms too much. This isn’t happening all the time, but it seems to be occurring more and more frequently. To be honest, it’s completely exhausting and I realized I need some help. I’ve also been getting these occasional rashes on my skin, though I don’t suppose that’s related.”

Step 1: Initial Information Gathering

In your small team, perform some research to gather information that might explain Xirui’s condition. Record any relevant findings.

- What are some different conditions that Xirui might be suffering from?
- What additional information might help lead you to a diagnosis?

Case 1: Xirui



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You notice that some of Xirui's joints indeed appear a bit inflamed and you order some tests. Below are the most relevant results:

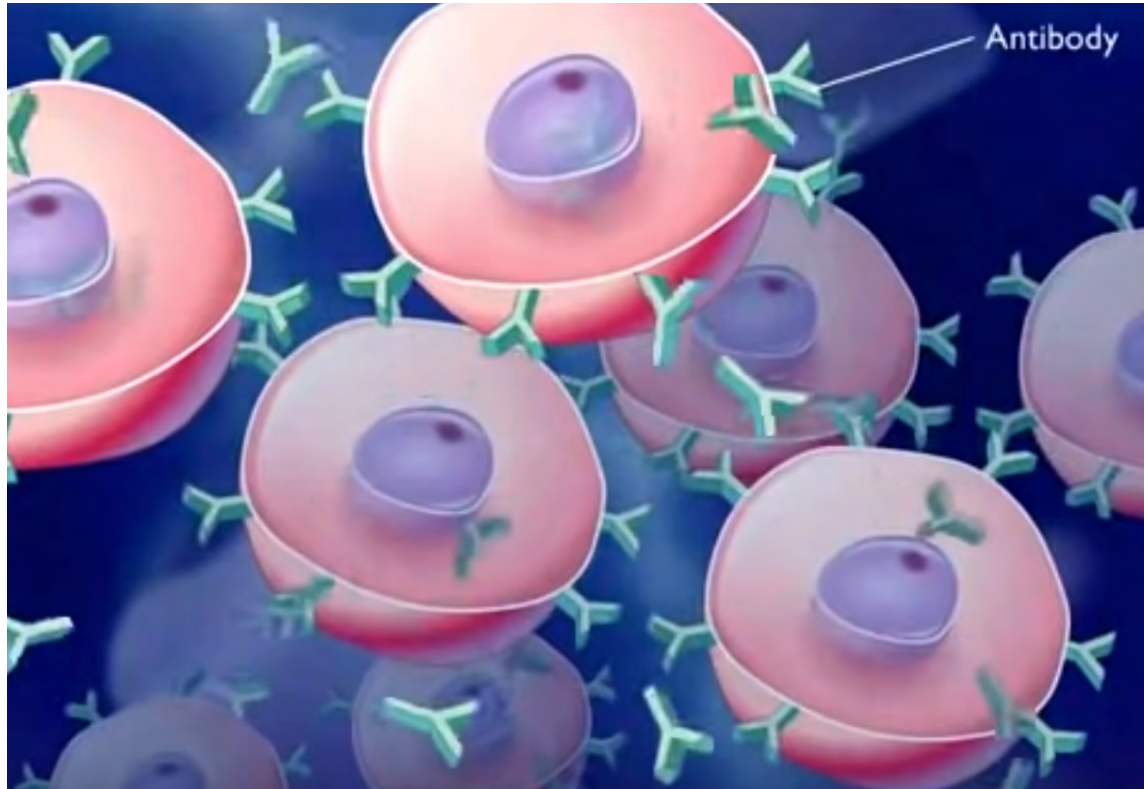
Blood Test	Xirui's Results
CBC	RBC: Low Platelets: Low
ANA	Positive

Step 2: Diagnosis

Perform additional research to interpret these results.

- What condition does Xirui have?
- What is the "ANA" test measuring?
- What does a positive test result for ANA tell us about what's happening in this disease?

Introducing (or reviewing) Adaptive Immunity



<https://www.youtube.com/watch?v=AucZlvEv29Y>

Case 1: Xirui

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CBC	RBC: Low Platelets: Low
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Step 3: Cellular Explanation

One of the hypothesized mechanisms for this disease is "clearance deficiency"

- Search for some images and information on "clearance deficiency"
- Make a list of the steps in clearance deficiency (some of the steps should be familiar from what we've learned about how the immune system works!)

efficient clearance
by phagocytes

apoptotic
cell

impaired clearance
secondary necrosis

NHD

SLE



ANTI-INFLAMMATION

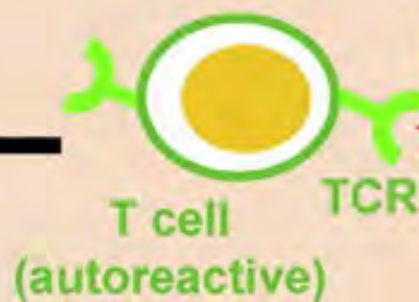
★ "find-me" and
● "eat-me" signals

INFLAMMATION



autoantibodies

humoral
AUTOIMMUNITY



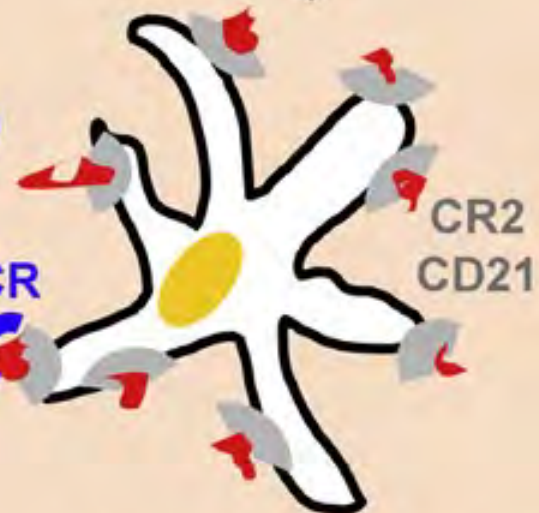
T cell
(autoreactive)

cellular
AUTOIMMUNITY

B cell
(autoreactive)

MHC

BCR



follicular dendritic
cell binding nuclear
autoantigens

Many of your B Cells are a tad auto-reactive
(react to your own body's proteins)

Helper T Cells Should Keep this in Check

B Cell Awaiting
Further
Stimulation

Helper T Cell Should Only
Activate B Cell if it's Not
Reacting to Body as Enemy



<https://youtu.be/rd2cf5hVaIM?t=386>